Taxonomy of Symbolic Structure and Autonomous Feedback

Distribution of the 46 included studies across two independent dimensions

Symbolic structure	Without autonomous feedback	With autonomous feedback
No symbolic structure	**Group A** Direct Generation 18 included studies (39.1%)	**Group D** Runtime Recovery 2 included studies (4.3%)
Lightweight or static symbolic structure	**Group B** Static Neuro-Symbolic 18 included studies (39.1%)	**Group D** Runtime Recovery 1 included studies (2.2%)
Formal symbolic structure	**Group C** Formal Verification 4 included studies (8.7%)	**Group D** Runtime Plan Adaptation and Recovery 3 included studies (6.5%)

Research gap: No included study demonstrates a complete closed-loop neuro-symbolic recovery cycle combining dynamic-world change, semantic diagnosis of physical failure, formal model updating, and autonomous replanning.